HW 3

Kevin Lin

10/15/2026

1

Let $S = \sum_{i=1}^n X_i$, and $S_\epsilon = \sum_{i=1}^n \epsilon X_i$. The Rademacher symmetrization lemma states that if $X_1, \ldots, X_n$ are independent mean zero random variables and $F : [0, \infty) \to \mathbb{R}$ is convex and increasing, then for $\epsilon_1, \ldots, \epsilon_n$ independent Rademacher random variables, we have:

$$E\left[F\left(\frac{1}{2}\left|\sum_{i=1}^n \epsilon_i X_i\right|\right)\right] \leq E\left[F\left(\left|\sum_{i=1}^n X_i\right|\right)\right] \leq E\left[F\left(2\left|\sum_{i=1}^n \epsilon_i X_i\right|\right)\right]$$

We will use the definition of a subgaussian norm:

$$||Y||_{\psi_2} = \inf\{K > 0 : E[\exp(Y^2/K^2)] \leq 2\}$$

First, define an upper bound. Let $K_\epsilon = ||S_\epsilon||_{\psi_2}$. Then, for any $K > K_\epsilon$, we have $E[\exp(S_\epsilon^2/K^2)] \leq 2$. Choose $F(t) = \exp(t^2/(2K)^2)$, which is convex and increasing. Then, by the Rademacher symmetrization lemma (right hand side), we have:

$$E\left[\exp\left(\frac{S^2}{(2K)^2}\right)\right] \leq E\left[\exp\left(\frac{(2|S_\epsilon|)^2}{(2K)^2}\right)\right] = E\left[\exp\left(\frac{S_\epsilon^2}{K^2}\right)\right] \leq 2$$

Then, by the definition of the subgaussian norm, we have $||S||_{\psi_2} \leq 2K$. Since this holds for every $K > K_\epsilon$, we have $||S||_{\psi_2} \leq 2||S_\epsilon||_{\psi_2}$, and we take $C = 2$.
Now, define a lower bound. Let $K = ||S||_{\psi_2}$. Then, for any $L > K$, we have $E[\exp(S^2/L^2)] \leq 2$. Choose $F(t) = \exp(t^2/L^2)$, which is convex and increasing. Then, by the Rademacher symmetrization lemma (left hand side), we have:

$$E\left[\exp\left(\frac{(1/2)|S_\epsilon|^2}{L^2}\right)\right] = E\left[\exp\left(\frac{|S_\epsilon|^2}{(2L)^2}\right)\right] \leq E\left[\exp\left(\frac{S^2}{L^2}\right)\right] \leq 2$$

Then, $||S_\epsilon||_{\psi_2} \leq 2L$. Since this holds for every $L > K$, we have $||S_\epsilon||_{\psi_2} \leq 2||S||_{\psi_2}$, which rearranges to $\frac{1}{2}||S_\epsilon||_{\psi_2} \leq ||S||_{\psi_2}$, and we take $c = 1/2$. Thus,

we can combine the two bounds to get:

$$\frac{1}{2}||S_\epsilon||_{\psi_2} \leq ||S||_{\psi_2} \leq 2||S_\epsilon||_{\psi_2}$$

$$\frac{1}{2}\left\|\sum_{i=1}^{n}\epsilon_i X_i\right\|_{\psi_2} \leq \left\|\sum_{i=1}^{n} X_i\right\|_{\psi_2} \leq 2\left\|\sum_{i=1}^{n}\epsilon_i X_i\right\|_{\psi_2}$$

where $c = 1/2$ and $C = 2$.

2